AI-Driven Markerless Kinematics in Pediatric Orthopedics: A Comprehensive Synthesis of Open-Source Frameworks and Clinical Deformity Analysis

The landscape of clinical gait analysis is undergoing a transformative shift as artificial intelligence and computer vision frameworks challenge the long-standing dominance of marker-based optoelectronic systems. In pediatric orthopedics, where patient compliance is often limited and the physical application of retroreflective markers can be cumbersome or distressing, markerless pose estimation offers a non-intrusive, scalable, and cost-effective alternative.[1] This report provides an exhaustive technical analysis of the state-of-the-art in markerless gait assessment, specifically tailored to the management of structural deformities such as genu varum/valgum, leg length discrepancy (LLD), and congenital talipes equinovarus (clubfoot). By synthesizing findings from recent clinical validations and exploring the underlying computational architectures of open-source libraries like MediaPipe Pose, OpenPose, and MMPose, this document outlines a technical roadmap for deep research and clinical implementation.

Computational Architectures of Markerless Pose Estimation

The efficacy of AI-driven gait analysis relies on the precision of human pose estimation (HPE) algorithms, which are broadly categorized into top-down and bottom-up methodologies. Understanding these mechanisms is fundamental for selecting the appropriate tech stack for specific orthopedic interventions.[4]

MediaPipe Pose and the BlazePose Pipeline

MediaPipe Pose, developed by Google, has emerged as a premier framework for real-time tracking on resource-constrained devices, including mobile phones and tablets.[6] The framework utilizes a top-down, two-stage pipeline consisting of a lightweight person detector followed by a pose landmarker model.[7] The detector first localizes the region of interest (ROI) by identifying facial and torso landmarks, which significantly reduces the computational search space for subsequent joint localization.[8]

What distinguishes MediaPipe Pose from traditional models is its comprehensive 33-landmark

topology, which extends beyond the 17 standard COCO keypoints to include detailed landmarks for the hands, feet, and face.[6] In clinical gait analysis, this extended set provides higher fidelity for lower-limb assessment, particularly at the ankle and foot segments.[10] The model outputs 3D Cartesian coordinates (x, y, z), where x and y are normalized to the image dimensions and z represents the estimated depth relative to the midpoint of the hips.[9]

OpenPose and Bottom-Up Association

OpenPose remains a foundational framework due to its pioneering use of Part Affinity Fields (PAFs) for multi-person 2D pose detection.[13] Unlike top-down models, OpenPose employs a bottom-up approach that identifies all joint candidates in an image simultaneously and then uses vector fields to associate joints with individual human skeletons.[4] This architecture is particularly robust in scenarios involving multiple subjects or significant self-occlusion, common in clinics where parents or therapists may be in the camera's view to support a pediatric patient.[4]

The OpenPose BODY_25 model is frequently cited in gait research for its inclusion of specialized foot keypoints, allowing for the calculation of the foot progression angle and metatarsophalangeal (MTP) joint dynamics.[15] Although computationally more demanding than MediaPipe, OpenPose's ability to handle complex spatial relationships makes it a preferred choice for high-precision laboratory settings using monocular or multi-camera setups.[16]

MMPose and Modular Deep Learning

MMPose, part of the OpenMMLab project, offers a highly modular PyTorch-based toolkit that allows researchers to implement and compare a wide variety of state-of-the-art (SOTA) models, including HRNet and RTMPose.[19] RTMPose, in particular, represents a breakthrough in top-down pose estimation, delivering unprecedented accuracy and real-time performance on edge devices.[6] Recent comparative studies have highlighted MMPose as the most effective framework for tracking lower-limb kinematics during walking tasks, demonstrating a high correlation $(r = 0.76–0.99)$ with 3D motion capture systems.[21]

Landmark Architecture	Landmark Count	Core Mechanism	Optimized Platform
MediaPipe Pose	33	Top-down (Detector-Tracker)	Mobile/Edge CPU

OpenPose	25 - 135	Bottom-up (PAF)	Desktop GPU
MMPose (RTMPose)	Customizable	Top-down/Transformer-based	Cross-platform
YOLO-Pose	17 (COCO)	One-stage (End-to-end)	Embedded Systems

Clinical Validation in Genu Varum and Valgum

The measurement of knee joint angles in the frontal and sagittal planes is critical for diagnosing and monitoring deformities such as bowlegs (genu varum) and knock-knees (genu valgum), which frequently arise from nutritional deficiencies like rickets.[4] Markerless systems must provide reliable data comparable to goniometry or optoelectronic gold standards to be viable in a clinical setting.[4]

Sagittal Plane Flexion and Extension

The analysis of knee flexion and extension during the gait cycle serves as a primary metric for assessing joint mobility and tracking the correction of bone bowing.[4] Validation studies utilizing MediaPipe Pose have demonstrated a high Pearson correlation ($r = 0.941$) with Kinovea, a standard marker-based 2D video analysis tool.[4] The mean absolute error (MAE) for knee angle measurement via MediaPipe Pose is approximately 5.88 degrees.[4]

Mathematically, the knee angle (θ_{knee}) is derived using the 3D coordinates of the hip (P_{hip}), knee (P_{knee}), and ankle (P_{ankle}). If we define vectors $\mathbf{u} = P_{hip} - P_{knee}$ and $\mathbf{v} = P_{ankle} - P_{knee}$, the angle is calculated as:

$$\theta_{knee} = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$$

This approach allows for the extraction of full range-of-motion (ROM) curves throughout the gait cycle, identifying specific points of peak flexion or extension that may indicate residual

stiffness or compensatory patterns.[15]

Frontal Plane Deformity and Dynamic Knee Valgus

Assessing genu valgum and varum requires frontal plane analysis to measure the excursion of the knee relative to the mechanical axis of the limb.[26] OpenPose has been rigorously validated for calculating the knee valgus angle during high-intensity tasks like drop vertical jumps, showing no significant difference in precision when compared to expert human-manual analysis ($MAE = 2.4$ degrees).[27]

However, frontal plane measurements are highly susceptible to camera parallax and lens distortion. Research indicates that if the observer (camera) is positioned more than 15 degrees from the knee perpendicular, the "true" angle can be significantly obscured, with perceived angles reaching up to 121 degrees for a true 90-degree flexion.[23] This highlights the necessity of standardized camera placement protocols—ideally at a height of 1000 mm and a distance of 2000 mm from the subject—to ensure inter-session reliability in clinical monitoring.[4]

Quantifying Asymmetry in Leg Length Discrepancy (LLD)

Leg length discrepancy (LLD) presents a unique challenge, as it forces the patient to adopt compensatory mechanisms such as unilateral pelvic tilt, equinus gait on the shorter side, or knee flexion on the longer side.[16] Markerless pose estimation allows for the objective quantification of these spatiotemporal and kinematic asymmetries without the labor-intensive setup of a traditional gait lab.[16]

Spatiotemporal Parameter Derivation

In moving toward decentralized clinical applications, researchers have developed versatile workflows that can accommodate various camera perspectives, such as sagittal (side) and frontal (front/back) views.[16] While sagittal views generally provide more accurate spatiotemporal data, frontal plane recordings are essential for analyzing pelvic tilt and hip abduction/adduction.[28]

To dimensionalize pixel distances in frontal-view videos, a trigonometric approach is employed to estimate depth-change based on the subject's pixel height. The size ratio (s_{ratio}) between a reference depth and an instantaneous depth allows for the estimation of depth change (Δd) as:

$$\Delta d = f \cdot \left(\frac{s_{Ref}}{s_i} - 1 \right)$$

where f is the camera's focal length, s_{Ref} is the subject's height in pixels at a known reference distance, and s_i is their current height.[16] This depth information is then used to calculate step length and gait speed on a stride-by-stride basis, revealing the discrete spatial asymmetries indicative of LLD.[16]

Pelvic and Trunk Kinematics

Compensatory pelvic tilt is a hallmark of LLD. Markerless HPE frameworks can track the medial-lateral and vertical trajectories of the hip landmarks (23 and 24 in MediaPipe) to derive the pelvic orientation.[24] While reliability for pelvic tilt (ICC = 0.41) and trunk tilt (ICC = 0.24) is lower than for joint angles, recent advances in body-centric elevation mapping and dense scene reconstruction (e.g., EgoHDM) are improving the stability of these global kinematic measures in real-world environments.[32]

Spatiotemporal Parameter	Mean Error (MAE)	Correlation (r)
Step Length (Sagittal)	0.02 m	0.99
Step Length (Frontal)	0.07 m	0.96
Step Time	0.01 s	Strong
Gait Speed	0.03 m/s	Strong
Stride Duration	2% Error	High

Markerless 3D Analysis for Clubfoot (Talipes

Equinovarus)

Clubfoot management relies on the Ponseti method of manipulation and casting, often followed by surgical tendon transfers to prevent dynamic recurrence.[34] Markerless 3D kinematics are particularly valuable here, as they allow for the longitudinal tracking of foot-ankle mechanics during and after the casting phase.[1]

Multi-Segment Foot Modeling with RGB-D Sensors

Standard HPE models often treat the foot as a single rigid segment, failing to capture the articulation of the metatarsophalangeal (MTP) joint, which is crucial for identifying pathological foot loading and supination in clubfoot.[1] To bridge this gap, researchers at Politecnico di Torino developed a 3D markerless protocol using single RGB-Depth (RGB-D) cameras like the Azure Kinect.[1]

This method involves creating a subject-specific 2D foot template from multiple static views, which is then mapped to a two-segment 3D foot model (mid-rear foot and forefoot).[3] The algorithm reconstructs 3D trajectories by matching the template to dynamic point clouds using the Iterative Closest Point (ICP) algorithm.[3] This enables the measurement of sagittal ankle and MTP kinematics with an angular offset of only 3.5 to 3.8 degrees compared to marker-based gold standards.[1]

Monitoring the Post-Casting Correction Process

Longitudinal gait analysis in children with clubfoot reveals specific success indicators, such as the normalization of peak ankle power at push-off and the reduction of internal foot progression angles.[39] Markerless systems facilitate frequent, low-cost assessments to detect early signs of "calcaneus gait" (excessive dorsiflexion) or residual "equinus" (limited dorsiflexion).[39] By integrating these findings into a clinical dashboard, surgeons can monitor the recovery of the MTP joint's role in foot loading and ensure that post-casting orthoses are effectively maintaining the correction.[3]

Integration with Biomechanical Modeling: The OpenSim Pipeline

To transform pose estimation landmarks into actionable clinical insights, it is necessary to move from surface-level tracking to biomechanical modeling. OpenSim serves as the primary open-source platform for this transition, enabling the calculation of joint moments and muscle forces.[42]

Pose2Sim: From OpenPose to OpenSim

Pose2Sim is an open-source Python package that provides a robust, unified workflow for 3D

markerless kinematics.[42] The framework integrates multiple calibrated video views to reconstruct 3D joint positions and then constrains these coordinates to a physically consistent OpenSim skeletal model.[42]

The Pose2Sim pipeline involves:

1. **Camera Calibration:** Utilizing checkerboards or QTM-based calibration data to establish the camera matrix and distortion coefficients.[42]
2. **3D Triangulation and Filtering:** Reconstructing 3D coordinates from multiple 2D views and applying Butterworth or median filters to remove noise.[42]
3. **Model Scaling:** Adjusting a generic musculoskeletal model to match the participant's specific limb lengths and segment masses.[46]
4. **Inverse Kinematics (IK):** Solving for joint angles that minimize the error between the triangulated markers and the model's virtual markers.[42]

This integration is particularly powerful for pediatric populations, as it allows clinicians to simulate the biomechanical effects of surgical modifications—such as an Achilles tenotomy in clubfoot—before actual intervention.[43]

Differentiable Musculoskeletal Models (Diff-MSM)

A nascent but high-potential area of research is the development of Differentiable MusculoSkeletal Models (Diff-MSM).[49] Unlike traditional IK, which treats the body as a set of rigid links, Diff-MSM uses automatic differentiation to simultaneously identify muscle and bone parameters from observable motion.[49] By training a deep neural network to integrate these physical constraints directly into the pose estimation process, researchers can resolve depth ambiguities and ensure that the estimated gait parameters are physiologically plausible.[50]

Open Source Link Matrix for Deep Research

The following categorized links provide the codebases, documentation, and data structures necessary to implement a professional-grade clinical gait analysis workflow.

I. Core Pose Estimation Frameworks

- **MediaPipe Pose Landmarker (Google):** The primary solution for real-time 33-landmark 3D pose extraction on mobile devices.
  - https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker
- **OpenPose (CMU):** The gold standard for multi-person bottom-up pose detection with Part Affinity Fields.
  - https://github.com/CMU-Perceptual-Computing-Lab/openpose
- **MMPose (OpenMMLab):** A comprehensive modular toolkit for PyTorch-based HPE, including RTMPose and HRNet.
  - https://github.com/open-mmlab/mmpose

- **ViTPose:** A high-precision transformer-based pose estimation model suitable for complex clinical tracking.
  - (https://github.com/ViTAE-Transformer/ViTPose)
- **DeepLabCut:** Markerless tracking for user-defined features, highly customizable for specific orthopedic landmarks.
  - (https://github.com/DeepLabCut/DeepLabCut)

II. Biomechanical Modeling and Clinical Workflows

- **Pose2Sim (Pagnon et al.):** The essential bridge for connecting pose estimation landmarks to the OpenSim biomechanical engine.
  - https://github.com/perfanalytics/pose2sim
- **OpenSim Python API:** Documentation and resources for running musculoskeletal simulations in a Python environment.
  - https://simtk.org/projects/opensim
- **GaitAnalysis-PoseEstimation (Stenum et al.):** The official MATLAB/Python workflow for the PLOS Digital Health 2024 study on multi-perspective gait analysis.
  - https://github.com/janstenum/GaitAnalysis-PoseEstimation
- **AutoGaitA:** An automated kinematic analysis toolbox capable of handling groups and longitudinal data.
  - https://github.com/mahan-hosseini/AutoGaitA
- **vGait:** A specialized tool for markerless motion data filtering, direction estimation, and step event detection.
  - (https://github.com/DSGZ-MotionLab/vGait)

III. Angle Estimation and Clinical Goniometers

- **Mediapipe2DAngle:** A specific implementation for calculating knee and hip joint angles from MediaPipe output files.
  - https://github.com/brunobedo/mediapipe2dangle
- **Goniometer-Mediapipe:** A test-driven Python architecture for serving real-time joint-angle measurements via a Flask API.
  - https://github.com/vaishakk/goniometer-mediapipe
- **MotionQuantification:** A dashboard system for extracting pose data and visualizing symmetry trends for rehabilitation tracking.
  - https://github.com/krmstrong322/MotionQuantification
- **Knee Bend Detection:** A real-time webcam system for quantifying knee joint angles during exercises.
  - https://github.com/zohairbadshah/Knee_bend_detection

IV. Specialized Research Datasets

- **Health&Gait (Zenodo):** Multimodal dataset featuring 1,564 videos of 398 participants with pose estimation and OptoGait ground truth.

    - https://doi.org/10.5281/zenodo.14039922
- **GaitRec (Figshare):** Large-scale ground reaction force dataset for patients with musculo-skeletal impairments at the hip, knee, and ankle.
    - https://doi.org/10.6084/m9.figshare.c.4788012
- **Typically Developing (TD) Children Gait Data:** Publicly shared treadmill biomechanics data for age-matched control comparisons.
    - (https://mcri.figshare.com/articles/dataset/Gait_dataset_-_Typically_developing_children_processed_with_CGM2/28216448)
- **Foot3D:** High-resolution foot scans and multiview images for evaluating 3D foot optimization and reconstruction.
    - (https://github.com/OllieBoyne/Foot3D)
- **AddBiomechanics Dataset:** A cloud-based platform for biomechanical data sharing and analysis.
    - https://addbiomechanics.org/

Technical Challenges and Optimization Strategies

Implementation of these frameworks in a clinical setting requires addressing systematic biases and environmental constraints that can compromise the validity of the orthopedic assessment.[4]

Temporal Smoothing and Signal Filtering

Pose estimation landmarks frequently exhibit high-frequency jitter, particularly in the depth (z) axis or under poor lighting.[9] Clinical standard workflows implement low-pass Butterworth filters to smooth joint trajectories.[10] A 4th-order Butterworth filter with a 5-6 Hz cutoff frequency is recommended for gait and sit-to-stand movements, as these behaviors have dominant frequency content below 5 Hz.[10]

In Python, this is implemented using scipy.signal.filtfilt() to achieve zero-phase forward-backward filtering, which prevents temporal shifts in gait event detection:

Python

```
from scipy.signal import butter, filtfilt

def apply_smoothing(data, cutoff=6, fs=30, order=4):
    nyquist = 0.5 * fs
    normal_cutoff = cutoff / nyquist
    b, a = butter(order, normal_cutoff, btype='low', analog=False)
```

```
    return filtfilt(b, a, data)
```

Gap Filling and Outlier Correction

Occlusions—such as a pediatric patient's arm crossing their leg during walking—can lead to missing landmarks.[9] Systems like "vailá" and the Stenum workflow employ linear or cubic spline interpolation for gaps spanning up to 0.12 seconds.[9] For larger gaps, more advanced "pose tracking" logic is required to distinguish between left and right limbs and prevent limb-swapping errors.[16]

Resolution of Depth Ambiguity

Monocular 3D HPE is an inherently ill-posed problem because different 3D poses can project to identical 2D images.[55] To mitigate this in orthopedic screening, practitioners are encouraged to use a multi-view approach, where two sets of 2D coordinates from different smartphone angles are matched through convolutional networks or direct triangulation to reconstruct true 3D space.[42] If only a single camera is available, adding a physical constraint—such as ensuring the subject's center of mass stays within their supporting feet—can significantly resolve depth inaccuracies.[55]

Future Horizons: State-Space Models and Foundation Gait Models

The next generation of AI-driven gait analysis is moving toward sequence modeling architectures that overcome the quadratic complexity of traditional Transformers.[52]

Pose3DM and Mamba Architectures

The "Pose 3D Human Estimation from Mamba" (Pose3DM) framework represents a paradigm shift in real-time gait assessment.[52] By utilizing bidirectional Mamba blocks, this model captures complex spatiotemporal dependencies with linear complexity, enabling the analysis of long video sequences with minimal memory latency.[52] For clinicians, this means the ability to process continuous, minutes-long walking bouts in real-time during a consultation, providing immediate feedback on deformity progression.[52]

GaitDynamics Foundation Models

"GaitDynamics" is a newly proposed generative foundation model for general-purpose gait dynamics prediction.[57] This model can estimate external ground reaction forces (GRF) from kinematic inputs and predict the influence of gait modifications on knee loading without the need for force plate experiments.[57] For orthopedic surgeons, this enables "experiment-free" simulation of how a specific brace or surgical correction will impact joint adduction moments,

allowing for highly personalized treatment planning.[57]

Conclusions and Practical Implementation Roadmap

The synthesis of AI frameworks and biomechanical modeling tools establishes a powerful infrastructure for the management of pediatric structural deformities. To satisfy the demands of expert-level research and clinical deployment, the following roadmap is proposed:

1. **Framework Selection:** Utilize **MediaPipe Pose** for rapid edge-based screening and telerehabilitation dashboards due to its 33-landmark topology and mobile optimization.[6] Implement **MMPose/RTMPose** for high-precision clinical research requiring superior lower-limb tracking accuracy.[21]
2. **Standardized Acquisition:** Implement strict camera protocols (perpendicular alignment, 2m distance) to minimize Parallax-induced angle overestimation.[4]
3. **Data Preprocessing Pipeline:** Integrate zero-phase **Butterworth filtering** and **spline interpolation** to ensure the integrity of the ROM waveforms and spatiotemporal asymmetry measures.[9]
4. **Biomechanical Enrichment:** Deploy the **Pose2Sim to OpenSim** workflow to calculate clinically actionable metrics like joint moments and muscle loading, moving beyond simple surface coordinates.[42]
5. **Longitudinal Monitoring:** Leverage **Mstudio** and **MotionQuantification** dashboards to track patient progress across multiple visits, providing objective data to validate post-casting recovery in clubfoot or deformity correction in rickets.[58]

By integrating these open-source resources and computational strategies, clinicians can provide more equitable, accessible, and high-precision care for children suffering from complex orthopedic and structural deformities.[1]

Goal	Primary Resource	Secondary Resource
Angle Extraction	(https://github.com/brunobedo/mediapipe2dangle)	Goniometer-Mediapipe
Biomechanics	(https://github.com/perfanalytics/pose2sim)	(https://simtk.org/projects/opensim)
Gait Asymmetry	(https://github.com/jansteum/GaitAnalysis-PoseEstimation)	vGait

Data Comparison	Health&Gait	(https://doi.org/10.6084/m9.figshare.c.4788012)
Visual Dashboard	MotionQuantification	(https://github.com/hunminkim98/MStudio)

Works cited

1. Markerless movement analysis based on RGB and Depth sensor Technology for clinical applications - Politecnico di Torino, accessed March 16, 2026, https://iris.polito.it/retrieve/dffd66b6-be49-41b5-9737-6ac9638cd282/conv_phd_thesis_final_dilettabalta.pdf
2. A Model-Based Markerless Protocol for Clinical Gait Analysis Based on a Single RGB-Depth Camera - IRIS - Politecnico di Torino, accessed March 16, 2026, https://iris.polito.it/retrieve/0c60c097-7ec7-4f6e-8e45-3f2224afa54d/A_Model-Based_Markerless_Protocol_for_Clinical_Gait_Analysis_Based_on_a_Single_RGB-Depth_Camera_Concurrent_Validation_on_Patients_With_Cerebral_Palsy.pdf
3. A preliminary validation of a 3D markerless method for estimating the kinematics of a two-segment foot model using a single RGB- - WebThesis - Politecnico di Torino, accessed March 16, 2026, https://webthesis.biblio.polito.it/27911/1/tesi.pdf
4. (PDF) Knee Flexion/Extension Angle Measurement for Gait Analysis Using Machine Learning Solution “MediaPipe Pose” and Its Comparison with Kinovea ® - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/369058814_Knee_FlexionExtension_Angle_Measurement_for_Gait_Analysis_Using_Machine_Learning_Solution_MediaPipe_Pose_and_Its_Comparison_with_Kinovea_R
5. Comparison of different pose estimation models for lower-body kinematics: A validation study | Scientific Journal of Sport and Performance, accessed March 16, 2026, https://sjsp.aearedo.es/index.php/sjsp/article/download/pose-estimation-models-validated-lower-body-kinematics/219/3811
6. Best Pose Estimation Models & How to Deploy Them, accessed March 16, 2026, https://blog.roboflow.com/best-pose-estimation-models/
7. Pose landmark detection guide | Google AI Edge, accessed March 16, 2026, https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker
8. Comparative and Improvement Study of 3D Human Pose Estimation Algorithms using Monocular Cameras - Preprints.org, accessed March 16, 2026, https://www.preprints.org/manuscript/202401.1668
9. Automated Gait Analysis Based on a Marker-Free Pose Estimation Model - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10384445/
10. Multimodal Motion Capture Toolbox for Enhanced Analysis of Intersegmental Coordination in Children with Cerebral Palsy and Typically Developing - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12922778/
11. MediaPipe-based extraction of joint ROM and position from patient videos for the

functional movement screen: An exploratory study - SOAR@USA, accessed March 16, 2026, https://soar.usa.edu/cgi/viewcontent.cgi?article=1097&context=phjpt
12. Eli-Bro/CV-Gait-Analysis-SU2023: A data acquisition tool ... - GitHub, accessed March 16, 2026, https://github.com/Eli-Bro/CV-Gait-Analysis-SU2023
13. A study on gait analysis based on the pose estimation algorithm RTMPose using knee angle as a metric - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/383252274_A_study_on_gait_analysis_based_on_the_pose_estimation_algorithm_RTMPose_using_knee_angle_as_a_metric
14. Verification of validity of gait analysis systems during treadmill walking and running using human pose tracking algorithm - Kyoto University, accessed March 16, 2026, https://repository.kulib.kyoto-u.ac.jp/bitstream/2433/261778/1/j.gaitpost.2021.02.006.pdf
15. OpenPose for 2D Gait Analysis: Knee Flexion Landmark Detection - GitHub, accessed March 16, 2026, https://github.com/batking24/OpenPose-for-2D-Gait-Analysis
16. Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change | PLOS Digital Health - Our journal portfolio - PLOS, accessed March 16, 2026, https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000467
17. Two-dimensional video-based analysis of human gait using pose estimation - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8099131/
18. (PDF) Predicting walkway spatiotemporal parameters using a markerless, pixel-based machine learning approach - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/393571592_Predicting_walkway_spatiotemporal_parameters_using_a_markerless_pixel-based_machine_learning_approach
19. Overview — MMPose 1.3.2 documentation, accessed March 16, 2026, https://mmpose.readthedocs.io/en/latest/overview.html
20. mmpose-0.29.0/README.md · camenduru/show at 33365020c067ef6fa051c355bc31fd56fe608e3b - Hugging Face, accessed March 16, 2026, https://huggingface.co/camenduru/show/blob/33365020c067ef6fa051c355bc31fd56fe608e3b/mmpose-0.29.0/README.md
21. Validation of Automated Countermovement Vertical Jump Analysis: Markerless Pose Estimation vs. 3D Marker-Based Motion Capture System - MDPI, accessed March 16, 2026, https://www.mdpi.com/1424-8220/24/20/6624
22. Predicting Body Movements for Person Identification under Different Walking Conditions | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/326777485_Predicting_Body_Movements_for_Person_Identification_under_Different_Walking_Conditions

23. Comparison of Wearable and Computer Vision Based Approaches to Knee Flexion Angle Measurement | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/372635483_Comparison_of_Wearable_and_Computer_Vision_Based_Approaches_to_Knee_Flexion_Angle_Measurement
24. Human Pose Estimation for Clinical Analysis of Gait Pathologies - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11097739/
25. Learning to Estimate Critical Gait Parameters from Single-View RGB Videos with Transformer-Based Attention Network, accessed March 16, 2026, https://arxiv.org/html/2312.00398v2
26. The effect of simulated marker misplacement on the interpretation of inter-limb differences during a change of direction task | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/347939777_The_effect_of_simulated_marker_misplacement_on_the_interpretation_of_inter-limb_differences_during_a_change_of_direction_task
27. Validity and Reliability of OpenPose-Based Motion Analysis in Measuring Knee Valgus during Drop Vertical Jump Test - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11366844/
28. Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10965062/
29. Accuracy of Video-Based Gait Analysis Using Pose Estimation During Treadmill Walking Versus Overground Walking in Persons After Stroke - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12142495/
30. Clinical gait analysis using video-based pose estimation: multiple perspectives, clinical populations, and measuring change - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/367972403_Clinical_gait_analysis_using_video-based_pose_estimation_multiple_perspectives_clinical_populations_and_measuring_change
31. Clinical gait analysis using video-based pose estimation: Multiple perspectives, clinical populations, and measuring change - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/379307775_Clinical_gait_analysis_using_video-based_pose_estimation_Multiple_perspectives_clinical_populations_and_measuring_change
32. 33 Pose landmarks in MediaPipe Pose Landmark Model - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/figure/Pose-landmarks-in-MediaPipe-Pose-Landmark-Model_fig1_369058814
33. EgoHDM: An Online Egocentric-Inertial Human Motion Capture, Localization, and Dense Mapping System - arXiv, accessed March 16, 2026, https://arxiv.org/html/2409.00343v2
34. Sport Ability during Walking Age in Clubfoot-Affected Children after Ponseti

Method: A Case-Series Study - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8001402/
35. Study Details | NCT03953430 | Gait Analysis in Children With Clubfoot Treated With Tibialis Anterior Tendon Transfer | ClinicalTrials.gov, accessed March 16, 2026, https://clinicaltrials.gov/study/NCT03953430
36. A preliminary validation of a 3D markerless method for estimating the kinematics of a two-segment foot model using a single RGB-D camera - WebThesis - Politecnico di Torino, accessed March 16, 2026, https://webthesis.biblio.polito.it/27911/
37. A comparative evaluation of two markerless methods for sagittal lower limb joint kinematics estimation based on a single RGB-D camera - WebThesis - Politecnico di Torino, accessed March 16, 2026, https://webthesis.biblio.polito.it/25847/1/tesi.pdf
38. (PDF) A Model-Based Markerless Protocol for Clinical Gait Analysis Based on a Single RGB-Depth Camera: Concurrent Validation on Patients With Cerebral Palsy - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/376349969_A_model-based_markerless_protocol_for_clinical_gait_analysis_based_on_a_single_RGB-depth_camera_concurrent_validation_on_patients_with_cerebral_palsy
39. This is a narrative review of the functional evaluation of clubfoot treatment with gait analysis, accessed March 16, 2026, https://atm.amegroups.org/article/view/69710/html
40. Gait Analysis after Initial Nonoperative Treatment for Clubfeet: Intermediate Term Followup at Age 5 - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2664431/
41. Efficacy of Clinical Gait Analysis: A Systematic Review - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/51196203_Efficacy_of_Clinical_Gait_Analysis_A_Systematic_Review
42. Pose2Sim: An open-source Python package for multiview markerless kinematics, accessed March 16, 2026, https://www.researchgate.net/publication/363559891_Pose2Sim_An_open-source_Python_package_for_multiview_markerless_kinematics
43. Diff-MSM: Differentiable MusculoSkeletal Model for Simultaneous Identification of Human Muscle and Bone Parameters | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/398059550_Diff-MSM_Differentiable_MusculoSkeletal_Model_for_Simultaneous_Identification_of_Human_Muscle_and_Bone_Parameters
44. davidpagnon/pose2sim: Markerless kinematics, from OpenPose to OpenSim - GitHub, accessed March 16, 2026, https://github.com/davidpagnon/pose2sim
45. perfanalytics/pose2sim: Markerless kinematics with any cameras — From 2D Pose estimation to 3D OpenSim motion - GitHub, accessed March 16, 2026, https://github.com/perfanalytics/pose2sim
46. qualisys/paf-opensim-example: Example for using OpenSim with the QTM Project

Automation Framework - GitHub, accessed March 16, 2026, https://github.com/qualisys/paf-opensim-example
47. Biomechanically Accurate Gait Analysis: A 3d Human Reconstruction Framework for Markerless Estimation of Gait Parameters - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/401523444_Biomechanically_Accurate_Gait_Analysis_A_3d_Human_Reconstruction_Framework_for_Markerless_Estimation_of_Gait_Parameters/download
48. Pose2Sim - Markerless kinematics, from OpenPose to OpenSim - Zenodo, accessed March 16, 2026, https://zenodo.org/records/14713203
49. Diff-MSM: Differentiable MusculoSkeletal Model for Simultaneous Identification of Human Muscle and Bone Parameters | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/394688730_Diff-MSM_Differentiable_MusculoSkeletal_Model_for_Simultaneous_Identification_of_Human_Muscle_and_Bone_Parameters
50. PyPose: A Library for Robot Learning with Physics-based Optimization - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/373308560_PyPose_A_Library_for_Robot_Learning_with_Physics-based_Optimization
51. Markerless Pixel-Based Pipeline for Quantifying 2D Lower Limb Kinematics During Squatting: A Preliminary Validation Study | Request PDF - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/398965656_Markerless_Pixel-Based_Pipeline_for_Quantifying_2D_Lower_Limb_Kinematics_During_Squatting_A_Preliminary_Validation_Study
52. Bidirectional Mamba-Enhanced 3D Human Pose Estimation for Accurate Clinical Gait Analysis - MDPI, accessed March 16, 2026, https://www.mdpi.com/2504-3110/9/9/603
53. A Dataset of Clinical Gait Signals with Wearable Sensors from Healthy, Neurological, and Orthopedic Cohorts - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12546693/
54. carolinaclemente00/3D-HPE-MediaPipe-Pose - GitHub, accessed March 16, 2026, https://github.com/carolinaclemente00/3D-HPE-MediaPipe-Pose
55. Human Pose Estimation Using MediaPipe Pose and Optimization Method Based on a Humanoid Model - ResearchGate, accessed March 16, 2026, https://www.researchgate.net/publication/368681318_Human_Pose_Estimation_Using_MediaPipe_Pose_and_Optimization_Method_Based_on_a_Humanoid_Model
56. Accuracy Evaluation of 3D Pose Reconstruction Algorithms Through Stereo Camera Information Fusion for Physical Exercises with MediaPipe Pose - PMC, accessed March 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11644880/
57. stanfordnmbl/GaitDynamics - GitHub, accessed March 16, 2026, https://github.com/stanfordnmbl/GaitDynamics
58. krmstrong322/MotionQuantification: Simple proof-of-concept app/scripts to show the simplicity of mediapipe's human pose estimation - GitHub, accessed

March 16, 2026, https://github.com/krmstrong322/MotionQuantification
59. hunminkim98/MStudio: A comprehensive toolkit for editing and managing 2D and 3D markers in motion capture and biomechanical studies. - GitHub, accessed March 16, 2026, https://github.com/hunminkim98/MStudio